

CE361: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

Pre-requisite courses:

- Introduction to computer and computer organization.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Process Management	06
3.	Process Synchronization	06
4.	Deadlocks	04
5.	Memory Management	10
6.	Secondary Storage and Input Output Management	04
7.	File Systems	04
8.	System Protection	03
9.	Unix/Linux File System	04
10.	Epilogue	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction	02 Hours	05%
	What is an OS? OS structure, System Calls, Functions of OS, Evolution Of OS, OS design consideration for Multiprocessor and Multicore		
2.	Process Management	06 Hours	13%
	Process: What is Process? Process Control Block, Process States, Process API: fork(), exec() and wait() system calls, Process Scheduling, Threads: Types of Threads, Multithreading model, Scheduler activation		

3.	Process Synchronization	06 Hours	13%
	Race Conditions, Critical Section, Peterson's Solution, hardware primitives for synchronization, Mutex Locks, Semaphores, Monitors and Classical problems in synchronization(Producer-Consumer, Reader's & Writer Problem, Dining Philosopher Problem).		
4.	Deadlocks	04 Hours	9%
	Modelling, Characterization, Prevention and Avoidance, Detection and Recovery		
5.	Memory Management	10 Hours	22%
	Address Space, Address Translation, Multiprogramming with Fixed and variable partitions, Swapping, Segmentation, Free Space Management, Paging: Principle Of Operation, Page Allocation, H/W Support For Paging. Multilevel paging, Virtual Memory: Concept, Performance of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets.		
6.	Secondary Storage and Input Output Management	04 Hours	09%
	I/O Devices: Device Controllers, Device Drivers, Direct Memory Access. Goals of The I/O S/W, Interrupt Handler, Device Independent I/O Software Disks: Disk architecture, Disks Scheduling Algorithm, RAID		
7.	File Systems	04 Hours	09%
	File: File Naming, File attributes, File Operations, File Structure, File Types, Access Methods, Directories: Structure, Pathnames, Directory Operations. File System Implementation: partitioning and mounting, Allocation methods: Contiguous Allocation, Linked Allocation, Indexed Allocation.		
8.	System Protection	03 Hours	07%
	Goals of protection, Principles of protection, Domain of protection, Access matrix, Implementation of Access matrix, Access control		
9.	Unix/Linux File System	04 Hours	09%
	Buffer Cache, Inodes, The system calls - ialloc, ifree, namei, alloc and free. Mounting and Unmounting , files systems, Network File systems. EXT file system in linux		
10.	Epilogue	02 hours	04%
	Pointers to advanced topics: distributed OS, multimedia OS, embedded OS, real-time OS, OS for multiprocessor machines, research problems in OS		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Analyze the basic structure of OS and architectural components involved in OS design
CO2	Evaluate the performance of various process scheduling algorithms after understanding various process states and its mechanism to manage
CO3	Evaluate the requirements of process synchronization and coordination in contemporary operating system.
CO4	Compare the various mechanisms to manage memory by applying memory management algorithms and schemes along with management of input/output devices
CO5	Compare the various mechanisms to manage files by applying file allocation methods along with security aspects
CO6	Analyze the administration of Linux operating system

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	2	-	-	-	-	-	-	1	-	-
CO2	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO3	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO4	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO5	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO6	2	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:❖ **Text book:**

1. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley
2. Operating Systems Internals and Design Principles, William Stallings, Seventh

❖ **Reference book:**

1. Operating Systems: Three Easy Pieces by Remzi Arpaci-Dusseau and Andrea Arpaci-Dusseau (available online for free)
2. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
3. Operating Systems, D.M. Dhamdhare, TMH
4. Unix System Concepts & Applications, Sumitabha Das, TMH
5. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

❖ **Web material:**

1. NPTEL course: <https://nptel.ac.in/courses/106/106/106106144/>
2. Simulator: <https://cpudemo.azurewebsites.net/#/>

❖ **Software:**

1. Simulator: OS Sim (OS Concepts Simulator)